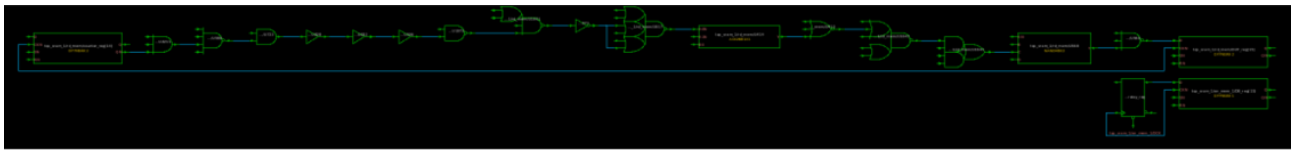
PDSD HW3 Guidelines

Dear All,

Please pay close attention to the following points regarding the preparation of HW3:

1. For “Hardware Performance evaluation”, we only consider the voting and median circuits from HW2.
2. For “Synthesis Process”, please include screenshots of each step's settings in *Design Compiler* in your report.
3. For “Circuit Analysis”, please exclusively use *Design Compiler* for discussion and provide screenshots for the relevant reports.
   I. Critical Path



   II. Area

```
****************************************
Report : area
Design : LEDDC
Version: M-2016.12
Date   : Thu Apr 14 01:57:36 2022
****************************************

Library(s) Used:

    slow (File: /mnt2/CBDK_IC_Contest_v2.1/SynopsysDC/db/slow.db)

Number of ports:                          919
Number of nets:                          4327
Number of cells:                         3145
Number of combinational cells:           2371
Number of sequential cells:               750
Number of macros/black boxes:               0
Number of buf/inv:                        398
Number of references:                       3

Combinational area:              30974.155122
Buf/Inv area:                     3131.702997
Noncombinational area:           14390.556938
Macro/Black Box area:                0.000000
Net Interconnect area:          320181.613251

Total cell area:                 45364.712060
Total area:                     365546.325310

***** End Of Report *****
```

   III. Power

```
****************************************

Library(s) Used:

    slow (File: /mnt2/CBDK_IC_Contest_v2.1/SynopsysDC/db/slow.db)

Operating Conditions: slow   Library: slow
Wire Load Model Mode: top

Design       Wire Load Model            Library
------------------------------------------------
LEDDC                tsmc13_wl10        slow

Global Operating Voltage = 1.08
Power-specific unit information :
    Voltage Units = 1V
    Capacitance Units = 1.000000pf
    Time Units = 1ns
    Dynamic Power Units = 1mW    (derived from V,C,T units)
    Leakage Power Units = 1pW


  Cell Internal Power  =   1.9461 mW   (92%)
  Net Switching Power  = 168.1870 uW    (8%)
                         ---------
Total Dynamic Power    =   2.1143 mW  (100%)

Cell Leakage Power     =  46.4185 uW


                 Internal         Switching           Leakage           Total
Power Group      Power            Power               Power             Power   (   %    )  Attrs
--------------------------------------------------------------------------------------------------
io_pad             0.0000           0.0000            0.0000            0.0000  (   0.00%)
memory             0.0000           0.0000            0.0000            0.0000  (   0.00%)
black_box          0.0000           0.0000            0.0000            0.0000  (   0.00%)
clock_network      0.0000           0.0000            0.0000            0.0000  (   0.00%)
register           1.9098        7.9062e-03        1.3958e+07            1.9317  (  89.40%)
sequential         0.0000           0.0000            0.0000            0.0000  (   0.00%)
combinational   3.6321e-02           0.1603        3.2460e+07            0.2291  (  10.60%)
--------------------------------------------------------------------------------------------------
Total              1.9461 mW        0.1682 mW     4.6419e+07 pW         2.1607 mW

***** End Of Report *****
```

IV. Delay

```
****************************************
Report : timing
         -path full
         -delay max
         -max_paths 1
         -sort_by group
Design : LEDDC
Version: M-2016.12
Date   : Thu Apr 14 01:59:32 2022
****************************************

Operating Conditions: slow   Library: slow
Wire Load Model Mode: top

  Startpoint: top_sram_1/wr_mem_1/DB_relay_reg
              (rising edge-triggered flip-flop clocked by DCK)
  Endpoint: top_sram_1/wr_mem_1/DB_reg[15]
            (falling edge-triggered flip-flop clocked by DCK)
  Path Group: DCK
  Path Type: max

  Des/Clust/Port     Wire Load Model       Library
  ------------------------------------------------
  LEDDC              tsmc13_wl10           slow

  Point                                                   Incr       Path
  --------------------------------------------------------------------------
  clock DCK (rise edge)                                   0.00       0.00
  clock network delay (ideal)                             1.00       1.00
  top_sram_1/wr_mem_1/DB_relay_reg/CK (DFFRX1)            0.00       1.00 r
  top_sram_1/wr_mem_1/DB_relay_reg/Q (DFFRX1)             0.56       1.56 f
  top_sram_1/wr_mem_1/DB_reg[15]/D (DFFNSRX1)             0.00       1.56 f
  data arrival time                                                  1.56

  clock DCK (fall edge)                                 650.00     650.00
  clock network delay (ideal)                             1.00     651.00
  clock uncertainty                                      -0.10     650.90
  top_sram_1/wr_mem_1/DB_reg[15]/CKN (DFFNSRX1)           0.00     650.90 f
  library setup time                                     -0.24     650.66
  data required time                                               650.66
  --------------------------------------------------------------------------
  data required time                                               650.66
  data arrival time                                                 -1.56
  --------------------------------------------------------------------------
  slack (MET)                                                      649.10
```

4. Please submit the synthesis-relevant files according to the example below:

*** All circuit designs, including port names, top module names, file names, and folder names, should follow the template provided earlier. ***

*** Submit the synthesis results of the adder circuit without overflow functionality and ensure that the pin assignments follow the template provided earlier. ***

I. Homework 1-1
   i. Folder name: RCA or CLA
   ii. File name: adder_syn.v (both for RCA or CLA)

II. Homework 2
   i. Folder name: voting or median
   ii. File name: voting_syn.v and median_syn.v

III. Folder Structure:

-------HW3_B083011056/
----------HW1-1/
----------RCA/
----------adder_syn.v
----------adder_syn.sdf
----------tb.v
----------CLA/
----------adder_syn.v
----------adder_syn.sdf
----------tb.v
----------HW2/
----------voting/

----------voting_syn.v
----------voting_syn.sdf
----------tb.v
----------median/
---------- median_syn.v
----------median_syn.sdf
----------tb.v
----------HW3_B083011056.pdf

5. The following link https://forms.gle/mcmh8D7pyuECbvLn6) is the hardware performance evaluation statistical form. Please fill in the form based on the time delay and area results generated by Design Compiler. If there are different versions, please select the best version for submission. This part will affect the assignment grades, so everyone must fill it out to protect your rights. Thank you.